RAJALAKSHMI ENGINEERING COLLEGE

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D2

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| s1 | | a | g | **g** | **t** | **a** | **b** |
| s2 | | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

// Helper function to find the maximum of two integers
int max(int a, int b) {
    return (a > b) ? a : b;
}

// Function to find the length of the Longest Common Subsequence
int longestCommonSubsequence(char* s1, char* s2) {
    int m = strlen(s1);
    int n = strlen(s2);

    // Using a 2D array for the DP table
    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            }
            else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    return dp[m][n];
}

int main() {
    // Arrays to hold the input strings (adjust size if your constraints are larger)
    char s1[1001];
    char s2[1001];

    // Read two strings from standard input
    if (scanf("%1000s", s1) == 1 && scanf("%1000s", s2) == 1) {
        // Output ONLY the integer result to match the expected format
        printf("%d\n", longestCommonSubsequence(s1, s2));
    }
```

```
44
45      return 0;
46  }
47
```

Check

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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